

Math 241, F12
Exam 2

Name: *Answer*
SSN: xxx-xx-

Instructions:

- There are totally 4 problems.
- You must show all of your work to receive a credit for a correct answer.
- No calculator is allowed in the exam.

Problem	Points	Score
1	24	
2	26	
3	24	
4	26	
Total	100	

Good Luck !

Problem 1. (24 points) Find the limit if it exists or show the limit does not exist.

(a) $\lim_{(x,y) \rightarrow (0,1/2)} e^{-x} \sin(\pi y)$

$$= e^0 \sin(\pi/2)$$

$$= 1$$

(b) $\lim_{(x,y) \rightarrow (1,0)} \ln \left(\frac{1+y^2}{x^2+xy} \right)$

$$= \ln \left(\frac{1+0^2}{1^2+1 \cdot 0} \right)$$

$$= \ln 1$$

$$= 0$$

(c) $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^3 + y^3}$

Let $f(x,y) = \frac{x^2 y}{x^3 + y^3}$

$$f(x,y) = \frac{x^3}{x^3 + x^3} = \frac{1}{2} \rightarrow \frac{1}{2} \text{ as } (x,y) \rightarrow (0,0) \text{ along } y=x$$

$$f(x,y) = \frac{0}{0+y^3} = 0 \rightarrow 0 \text{ as } (x,y) \rightarrow (0,0) \text{ along } x=0.$$

$$\Rightarrow \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^3 + y^3} \text{ does not exist.}$$

(d) $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^2 + y^2}$

$$= \lim_{(x,y) \rightarrow (0,0)} \underbrace{\left(\frac{x^2}{x^2 + y^2} \right)}_{\leq 1} (y) = 0 \rightarrow 0 \text{ as } y \rightarrow 0$$

Problem 2. (26 points)

(a) Use Chain Rule to find $\partial z / \partial s$ and $\partial z / \partial t$ when $z = \ln(x^2 + y)$, $x = s - t$, $y = \cos(st)$.

$$\begin{aligned}
 \text{Solution: } \frac{\partial z}{\partial s} &= \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s} \\
 &= \left(\frac{2x}{x^2 + y} \right) (1) + \left(\frac{1}{x^2 + y} \right) (-t \sin(st)) \\
 &= \frac{2(s-t) - t \sin(st)}{(s-t)^2 + \cos(st)} \\
 \frac{\partial z}{\partial t} &= \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t} \\
 &= \left(\frac{2x}{x^2 + y} \right) (-1) + \left(\frac{1}{x^2 + y} \right) (-s \sin(st)) \\
 &= \frac{-2(s-t) - s \sin(st)}{(s-t)^2 + \cos(st)}
 \end{aligned}$$

(b) Let $f(x, y, z) = \sqrt{xyz}$. Find the gradient $\nabla f(x, y, z)$ and its directional derivative at the point $(3, 2, 6)$ along the direction of the vector $\mathbf{v} = \langle -1, -2, 2 \rangle$.

$$\begin{aligned}
 \text{Solution: } \nabla f(x, y, z) &= \langle f_x, f_y, f_z \rangle \\
 &= \left\langle \frac{yz}{2\sqrt{xyz}}, \frac{xz}{2\sqrt{xyz}}, \frac{xy}{2\sqrt{xyz}} \right\rangle \\
 &= \left\langle \frac{1}{2} \sqrt{\frac{yz}{x}}, \frac{1}{2} \sqrt{\frac{xz}{y}}, \frac{1}{2} \sqrt{\frac{xy}{z}} \right\rangle \\
 \Rightarrow \nabla f(3, 2, 6) &= \left\langle \frac{1}{2} \sqrt{4}, \frac{1}{2} \sqrt{9}, \frac{1}{2} \sqrt{1} \right\rangle \\
 &= \left\langle 1, \frac{3}{2}, \frac{1}{2} \right\rangle \\
 \mathbf{v} = \langle -1, -2, 2 \rangle &\Rightarrow |\mathbf{v}| = \sqrt{9} = 3 \\
 \Rightarrow \mathbf{u} = \frac{\mathbf{v}}{|\mathbf{v}|} &= \left\langle -\frac{1}{3}, -\frac{2}{3}, \frac{2}{3} \right\rangle \\
 \Rightarrow \text{The directional derivative is} \\
 \nabla f(3, 2, 6) \cdot \mathbf{u} &= \left\langle 1, \frac{3}{2}, \frac{1}{2} \right\rangle \cdot \left\langle -\frac{1}{3}, -\frac{2}{3}, \frac{2}{3} \right\rangle \\
 &= -\frac{1}{3} - 1 + \frac{1}{3} \\
 &= -1
 \end{aligned}$$

Problem 3. (24 points)

(a) Find the local maximum and minimum values and saddle points of the function $f(x, y) = x^3 + y^3 - 3xy + 4$.

Solution: $f_x(x, y) = 3x^2 - 3y$ and $f_y(x, y) = 3y^2 - 3x$

$$\text{Solve } \begin{cases} 3x^2 - 3y = 0 \\ 3y^2 - 3x = 0 \end{cases} \Rightarrow \begin{cases} x^2 - y = 0 \Rightarrow y = x^2 \\ y^2 - x = 0 \end{cases} \Rightarrow (x^2)^2 - x = 0$$

$$\cdot x^4 - x = 0$$

$$x(x^3 - 1) = 0 \Rightarrow x(x-1)(x^2+x+1) = 0$$

critical points: (0,0) and (1,1).

$$\Rightarrow \begin{matrix} x=0 & \text{or} & x=1 \\ \downarrow & & \downarrow \\ y=0 & & y=1 \end{matrix}$$

$$f_{xx} = 6x, \quad f_{yy} = 6y, \quad f_{xy} = -3$$

$$\Rightarrow D(x, y) = (6x)(6y) - (-3)^2 = 36xy - 9$$

For (0,0), $D(0,0) = 0 - 9 < 0$, so it is a saddle point.

For (1,1), $D(1,1) = 36 - 9 = 27 > 0$, $f_{xx}(1,1) = 6 > 0$,
so $f(1,1) = 1 + 1 - 3 + 4 = 3$ is a minimum value.

(b) Use the method of Lagrange multipliers to find the maximum and minimum value of the function $f(x, y) = 4x + 6y$ subject to the constraint $x^2 + y^2 = 13$.

Solution: Let $g(x, y) = x^2 + y^2$. We need solve (x, y, λ) from

$$\begin{cases} \nabla f(x, y) = \lambda \nabla g(x, y) \\ g(x, y) = 13 \end{cases} \Rightarrow \begin{cases} f_x(x, y) = \lambda g_x(x, y) \\ f_y(x, y) = \lambda g_y(x, y) \\ g(x, y) = 13 \end{cases}$$

$$\Rightarrow \begin{cases} 4 = \lambda(2x) \Rightarrow x = \frac{2}{\lambda} \\ 6 = \lambda(2y) \Rightarrow y = \frac{3}{\lambda} \\ x^2 + y^2 = 13 \end{cases} \Rightarrow \left(\frac{2}{\lambda}\right)^2 + \left(\frac{3}{\lambda}\right)^2 = 13$$

$$\frac{13}{\lambda^2} = 13 \Rightarrow \lambda^2 = 1 \Rightarrow \lambda = \pm 1$$

i) $\lambda = 1$, then $x = 2, y = 3$. $f(2,3) = 26$

ii) $\lambda = -1$, then $x = -2, y = -3$. $f(-2,-3) = -26$

$\Rightarrow$ The maximum value is 26 and the minimum value is -26

Problem 4. (26 points)

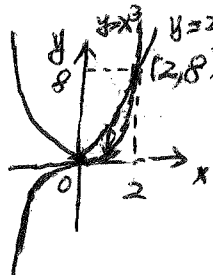
(a) Calculate the iterated integral $\int_0^1 \int_0^x \frac{y}{x^3+1} dy dx$

Solution:

$$\begin{aligned} & \int_0^1 \int_0^x \frac{y}{x^3+1} dy dx \\ &= \int_0^1 \left[\frac{1}{2} \frac{y^2}{x^3+1} \right]_{y=0}^{y=x} dx \\ &= \int_0^1 \frac{1}{2} \frac{x^2}{x^3+1} dx \\ &= \left[\frac{1}{6} \ln(x^3+1) \right]_{x=0}^{x=1} \\ &= \frac{1}{6} \ln 2 - \frac{1}{6} \ln 1 \\ &= \frac{1}{6} \ln 2 \end{aligned}$$

(b) Let D be the region bounded by $y = 2x^2$ and $y = x^3$. Find the volume of the solid under the surface $z = 2x + y$ and above the region D .

Solution:



$$\begin{aligned} & \begin{cases} y = 2x^2 \\ y = x^3 \end{cases} \Rightarrow x^3 - 2x^2 = 0 \\ & \quad \quad \quad x^2(x-2) = 0 \\ & \quad \quad \quad x = 0 \text{ or } x = 2 \\ & \quad \quad \quad \downarrow \quad \quad \downarrow \\ & \quad \quad \quad y = 0 \quad y = 8 \end{aligned}$$

$$\begin{aligned} V &= \int_0^2 \int_{x^3}^{2x^2} (2x + y) dy dx \\ &= \int_0^2 \left[2xy + \frac{y^2}{2} \right]_{y=x^3}^{y=2x^2} dx \\ &= \int_0^2 \left(-\frac{x^6}{2} + 4x^3 \right) dx \\ &= \left[-\frac{x^7}{14} + x^4 \right]_{x=0}^{x=2} \\ &= -\frac{128}{14} + 16 \\ &= \frac{48}{7} \end{aligned}$$